- *It should provide meaningful feedback.* If a test fails, a developer should be able to understand what the underlying requirement is trying to achieve, and also how it's interacting with the application to do so.
- *It should be reliable.* To get value out of the tests, the team needs to be able to trust the test results. A test should pass if (and only if) the application satisfies the underlying business requirement, and should fail if it doesn't. If a test does break for technical reasons, the issue should be easy to isolate and simple to fix. Although this sounds obvious, it can require care and discipline to ensure that the more complex tests respect this rule.
- *It should be easy to maintain.* A test that breaks too often when the application is updated or modified, or that requires constant maintenance to keep up to date, rapidly becomes a liability for the development team. If this happens too frequently, the developers will often simply cease updating the tests and leave them in a broken state. When this happens, any useful feedback from the test results is lost, and the time invested in building up the test suite is wasted.

In the rest of this chapter, we'll look at how you can write high-quality tests that respect these rules, and that provide valuable feedback and are robust and easy to maintain.

7.2 Automating your test setup process

Before you can automate any acceptance criteria, you need to make sure the system is in a correct and well-known initial state. Also, many automated acceptance criteria—particularly the end-to-end variety—will need to refer to or update data in a database. Some tests may need other services or resources, such as file systems or remote web services, to be initialized and configured. In order to have effective automated acceptance criteria, you need to be able to set up all of these things quickly, precisely, and automatically (see figure 7.4).

If your automated acceptance criteria use a database at all, you should reserve a dedicated database instance for them. Try to ensure that the database configuration is as close to the production one as possible. It's also a bad idea to use a database that's



Figure 7.4 Automated data setup is a key part of automated acceptance criteria.